Open Source & Tech Challenge

A Medium-Level Round on Git, Linux, GitHub & Security

LEVEL: MEDIUM

Participant: _____ **Date:** _____ **Score:** _____ / 100 pts

Medium: +20 for correct | -10 for wrong

PART 1 THINK CAREFULLY (GIT & LINUX)

Question 1: Git — Think Carefully

You have a Git repository with this history:

```
A → B → C
```

You create a new branch at C:

```
git branch feature
```

Then you make **2 commits on feature**. After that, you switch back to main.

How many commits does main have compared to feature?

- ☐ **A.** `main` has 2 more commits
- ☐ **B.** `feature` has 2 more commits
- ☐ **C.** Both have the same number of commits
- ☐ **D.** The commits disappear when switching branches

Question 2: Linux — Who Has Access?

A file has the following permissions:

```
-rw-r-----
```

The file owner is **Alice** and the group is **developers**. Bob is a member of the developers group. Charlie is neither the owner nor a member of the group.

Who can modify the file?

- ☐ **A.** Alice only
- ☐ **B.** Alice and Bob
- ☐ **C.** Alice, Bob and Charlie
- ☐ **D.** Bob only

PART 2 GITHUB CONTRIBUTION WORKFLOW

Question 3: You find a bug in an open-source project on GitHub. Which sequence makes the most sense if you want to contribute a fix?

- ☐ **A.** `Fork → Clone → Modify → Commit → Push → Pull Request`
- ☐ **B.** `Clone → Delete → Push → Fork → Pull Request`
- ☐ **C.** `Fork → Pull Request → Modify → Clone → Delete`
- ☐ **D.** `Push → Fork → Clone → Commit → Delete`

PART 3 CORE CONCEPTS (MULTIPLE CHOICE)

Question 4: What does CI/CD stand for in modern software engineering deployment pipelines?

- ☐ **A.** Code Integration / Code Debug
- ☐ **B.** Continuous Integration / Delivery
- ☐ **C.** Compiled Input / Core Database
- ☐ **D.** Central Instance / Cloud Data

Question 5: Which cryptographic protocol secures encrypted communication over computer networks (HTTPS)?

- ☐ **A.** FTP Protocol
- ☐ **B.** SSH Protocol
- ☐ **C.** TLS / SSL
- ☐ **D.** Telnet Service

KEY: 1. B (feature +2) 2. A (Alice only) 3. A 4. B (CI/CD) 5. C (TLS / SSL)